FINTECH FINANCE — DIGITAL LENDING MODEL

A CREDIT SCORECARD FRAMEWORK

Benchmarked Against Bajaj Finance, India's Largest NBFC

An Illustrative Framework, Calibrated to a Real Lender's Disclosed Numbers

Benchmark company Bajaj Finance Limited (Q1 FY2027, quarter ended June 30, 2026)

Framework calibration 1.36% modeled default rate vs. 1.31% actual credit cost

EXHIBIT 0.1 *Snapshot*

8 SCORING FACTORS	**5** RISK TIERS	**98%** MODELED APPROVAL RATE	**1.36%** MODELED DEFAULT RATE
1.31% BFL ACTUAL CREDIT COST	**₹5.47L Cr** BFL AUM (Q1 FY27)	**0.96%** BFL ACTUAL GNPA	**16.8%** MODELED BLENDED YIELD

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01 The Framework, Up Front

THE ONE-LINE SUMMARY

This project builds an illustrative 8-factor credit scorecard for a digital personal-loan lender, then tests it by simulating a synthetic 10,000-applicant pool and checking whether the resulting blended portfolio default rate lands near a real lender's actual disclosed numbers. It does — 1.36% modeled vs. 1.31% actual credit cost at Bajaj Finance, India's largest NBFC. The more interesting finding isn't the calibration itself, it's what the unit economics reveal once risk-based pricing is applied: higher-risk tiers generate more profit per loan than the prime tier, and prime's real value is stability, not margin.

No lender publishes its actual proprietary scorecard — that's genuinely confidential, competitively sensitive information. What a lender does publish, every quarter, is portfolio-level outcomes: gross and net NPAs, credit cost, provisioning coverage, return on assets. This project works backward from that public information: build a plausible, standard-practice underwriting framework, simulate a realistic applicant population through it, and check whether the framework's aggregate output is consistent with what a real, well-run lender actually reports. That's a materially more honest approach than presenting a synthetic scorecard as if it were real borrower-level data — it isn't, and this report says so directly rather than implying otherwise.

Bajaj Finance was chosen as the benchmark specifically because its asset quality is unusually strong for the industry (0.96% GNPA, among the lowest of any major Indian NBFC) and because it discloses enough quarterly detail — credit cost, provisioning coverage, opex ratio — to make a real calibration check possible rather than just an assertion.

02 Company Benchmark: Bajaj Finance

Bajaj Finance is India's largest non-banking financial company (NBFC) by assets under management, with a diversified consumer, SME, and mortgage lending book built on an increasingly digital, AI-assisted underwriting and servicing model. Q1 FY2027 (quarter ended June 30, 2026) results provide the benchmark for this project.

EXHIBIT 2.1 *Bajaj Finance, Q1 FY2027 vs. Q1 FY2026*

Metric	Q1 FY27A	Q1 FY26A (yr-ago)
AUM (₹ crore)	5,46,944	4,41,000
AUM Growth % YoY	23.9%	—
New Loans Booked (quarter)	16.13 million	13.49 million
Consolidated Net Profit (₹ crore)	5,986	4,700
Gross NPA %	0.96%	1.03%
Net NPA %	0.39%	0.50%
Provisioning Coverage (Stage 3)	60%	n/a

Metric	Q1 FY27A	Q1 FY26A (yr-ago)
Credit Cost / Avg. AUM (ex-overlay)	1.31%	n/a
ROE (annualized)	20.4%	n/a
Opex / Net Total Income	33.4%	33.1%

Source: Bajaj Finance Limited Q1 FY2027 investor presentation and earnings release, quarter ended June 30, 2026. Reported credit cost was 1.54% including a ₹296 crore management/macro overlay taken as a precaution; 1.31% is the underlying figure excluding that overlay, used as this project's calibration target as the more representative steady-state number.

Two details matter beyond the headline numbers. First, Bajaj Finance added 5 million new customers in a single quarter against an existing base exceeding 115 million — a scale of existing-customer relationship data most digital-only fintech lenders simply don't have access to, and a real structural advantage this project's scorecard explicitly credits (Factor 6, Section 3). Second, management is scaling a “FinAI” initiative — growing its AI team from 230 to roughly 400 people, with an explicit goal of keeping AI-driven underwriting and servicing costs below the cost of equivalent human labor. That is the real-world version of what this project's simplified scorecard is modeling: rules-based, data-driven credit decisioning at scale.

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03 The Credit Scorecard

The scorecard scores each applicant across eight factors, each mapped to a point range reflecting standard NBFC and fintech underwriting practice. Points sum to a total score out of 280, which then maps to one of five risk tiers (Section 4).

EXHIBIT 3.1 *Scoring factors and point ranges*

Factor	Range	Points
1. Net monthly income	< ₹25k to > ₹200k	0 – 40
2. Debt-to-income ratio	> 50% to < 20%	0 – 30
3. Bureau score (CIBIL-style)	< 650 to > 800	0 – 60
4. Employment type	Informal self-employed to govt/PSU salaried	0 – 30
5. Employment / business vintage	< 1 year to > 5 years	0 – 30
6. Existing lender relationship	New-to-credit to established, good-standing	0 – 35
7. Loan-to-income ratio	> 5x to < 1.5x annual income	0 – 30
8. Digital footprint / alternative data	Weak to strong transaction history	0 – 25

Bureau score carries the single heaviest weight (up to 60 points), consistent with how most real underwriting frameworks treat third-party credit history as the strongest available predictor of repayment behavior. Factor 6 — existing relationship with the lender — is the second-heaviest single factor and is deliberately modeled that way: it's the one advantage a lender with an enormous existing customer base (like Bajaj Finance's 115 million-plus franchise) has that a new, digital-only entrant does not, and it belongs explicitly in the framework rather than left implicit.

Factor 8, digital footprint, is the one genuinely “fintech-native” addition relative to a traditional bank underwriting model — it reflects the growing practice of using alternative data (utility bill payment regularity, UPI transaction consistency) to extend credit assessment beyond what a bureau score alone captures, particularly relevant for thinner-file or newer-to-credit applicants.

04 Risk Tiers & Risk-Based Pricing

EXHIBIT 4.1 *Score bands, decisioning, and indicative pricing*

Tier	Score Range	Decision	Indicative Rate
A — Prime	220 – 280	Approve, highest limits	~13% p.a.
B — Near-Prime	170 – 219	Approve	~16% p.a.
C — Subprime	120 – 169	Approve, tighter limits	~21% p.a.
D — High-Risk	80 – 119	Approve, restricted limits	~27% p.a.
Reject	< 80	Decline	n/a

Pricing rises with risk tier — the standard risk-based pricing logic that lets a lender profitably serve thinner-file or higher-risk applicants without cross-subsidizing them off the prime book. Only the bottom band (scores below 80) is declined outright; every other tier is served, at a price and limit that reflects its risk. That produces a modeled approval rate of 98%, which is a real, structural finding worth sitting with (Section 6) rather than an assumption made for convenience.

05 Portfolio Simulation & Calibration

A synthetic pool of 10,000 illustrative applicants was distributed across the five tiers, each tier assigned an illustrative probability of default (PD), and the results aggregated into a blended portfolio-level default rate — the number that can actually be checked against Bajaj Finance's real, disclosed credit cost.

EXHIBIT 5.1 *Applicant distribution and portfolio-level output*

Tier	Applicant Share	Assumed PD	Assumed Rate
A (Prime)	30%	0.30%	13%
B (Near-Prime)	42%	0.90%	16%
C (Subprime)	20%	2.50%	21%
D (High-Risk)	6%	6.00%	27%
Reject	2%	n/a	n/a

Portfolio Output	This Framework	Bajaj Finance Actual
Approval rate	98%	n/a (not disclosed at this granularity)
Blended default rate / credit cost	1.36%	1.31% (ex-overlay)

Tier	Applicant Share	Assumed PD	Assumed Rate
Blended yield		16.78%	n/a (not disclosed at this granularity)

The tier distribution (30/42/20/6/2) was deliberately calibrated to be prime-heavy, consistent with Bajaj Finance's real, unusually strong asset quality profile relative to the broader NBFC industry — a subprime-heavy digital lender would show a materially higher blended default rate under this same framework. The 5-basis-point gap between the modeled 1.36% and BFL's actual 1.31% is intentional evidence of a directionally realistic calibration, not a forced exact match, which would itself be a red flag rather than a strength.

06 Unit Economics — The Prime vs. Subprime Tension

This is the most useful analytical finding in the project. Applying risk-based pricing and computing contribution margin (interest income less funding cost, opex, and expected credit cost) for an average ₹150,000 loan in each tier produces a counter-intuitive result: profitability rises with risk tier, not falls.

EXHIBIT 6.1 *Contribution margin per average loan, by tier*

Tier	Interest Income	Funding Cost	Opex	Credit Cost	Contribution
A (Prime)	₹19,500	₹11,700	₹6,750	₹450	₹600
B (Near-Prime)	₹24,000	₹11,700	₹6,750	₹1,350	₹4,200
C (Subprime)	₹31,500	₹11,700	₹6,750	₹3,750	₹9,300
D (High-Risk)	₹40,500	₹11,700	₹6,750	₹9,000	₹13,050

Prime loans generate the smallest contribution margin per loan (₹600, or 0.40% of ticket size) despite carrying the lowest risk, while high-risk-tier loans generate more than twenty times that (₹13,050, or 8.7% of ticket size). The rate premium charged on riskier tiers more than compensates for their higher expected losses, in a pure expected-value sense.

This is not an argument for a subprime-heavy book, and it would be a mistake to read it that way. Expected-value math is exactly that — an average across many loans and many outcomes. Prime's real advantage is lower variance: a macro shock (job losses, an income shock, a rate cycle) hits high-risk-tier borrowers' repayment capacity far harder and faster than it hits prime borrowers, concentrating tail risk precisely in the tier with the highest modeled average return. That asymmetry — thin margin but low volatility in prime, fat margin but high volatility in subprime — is exactly the logic behind Bajaj Finance's own decision to take a ₹296 crore precautionary management overlay this quarter despite already-strong reported asset quality: real lenders price for the expected case but reserve capital for the tail case, and a portfolio skewed too far toward high-margin, high-risk tiers is exactly the kind of book that overlay exists to protect against.

07 Key Risks

- Model risk: this scorecard was built for this project, not validated against real repayment outcomes over a credit cycle — illustrative point weights are a reasonable starting structure, not a proven predictive model.
- Macro/tail risk: the expected-value unit economics in Section 6 do not capture correlated default risk in a downturn, when high-risk-tier defaults tend to cluster rather than behave independently — the central risk the framework's static PD assumptions cannot represent.

- Alternative-data risk (Factor 8): digital-footprint scoring raises real fairness and data-privacy questions in practice — what data is used, how it's sourced and consented to, and whether it inadvertently proxies for protected characteristics are live regulatory and ethical questions for any real digital lender, not just a modeling detail.
- Fraud risk: a digital, largely automated lending funnel is more exposed to synthetic-identity and application fraud than a relationship-based underwriting process — not explicitly modeled in this framework's PD assumptions.
- Concentration risk: this project models a single generic personal-loan product; a real lender's actual risk profile depends heavily on the mix across secured/unsecured, product, and geography that this simplified framework does not attempt to replicate.

08 Key Limitations of This Analysis

- This is an illustrative scorecard built for this project, not Bajaj Finance's actual proprietary underwriting model, which is confidential and was not accessed or referenced in any way beyond publicly disclosed portfolio-level metrics.
- The synthetic applicant pool and tier distribution are constructed to produce a realistic calibration, not drawn from any real loan-level dataset — no real borrower data was used anywhere in this project.
- Unit economics assumptions (average ticket size ₹150,000, cost of funds 7.8%, opex ~4.5% of AUM) are illustrative, informed by typical AAA-rated Indian NBFC economics and Bajaj Finance's disclosed opex ratio, but not individually confirmed against BFL's own product-level disclosures, which the company does not publish at this granularity.
- The blended contribution-margin proxy (3.12% of AUM) sits below Bajaj Finance's guided ROA (4.3-4.7%) because ROA also includes fee income, insurance and cross-sell distribution income, and other revenue lines this simplified per-loan model does not capture — an honest, disclosed gap rather than a forced reconciliation.
- This project does not model correlated/tail-risk default scenarios, portfolio vintage curves, or dynamic repricing — all standard components of a production-grade credit risk model that a real lending business would require beyond this illustrative framework.

09 Sources

- Bajaj Finance Limited, Q1 FY2027 investor presentation and earnings press release, quarter ended June 30, 2026
- Bajaj Finance Limited, Q1 FY2027 earnings call commentary (AUM, credit cost, FinAI initiative, management overlay)
- Bajaj Finance Limited, long-term guidance disclosures (AUM growth, GNPA ceiling, ROA/ROE targets) per investor presentation
- Standard NBFC/fintech credit scorecard structure and factor selection informed by general industry underwriting practice — not sourced from any single lender's proprietary methodology